

Foundation or Formula? A Simulation-Based Comparison of Google's TimesFM-3 and Classical Time-Series Models

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Abstract

Time-series foundation models such as Google's TimesFM-3, a 330-million-parameter model released in August 2026, forecast series they were never trained on. Whether such models should replace classical methods such as ARIMA and exponential smoothing cannot be settled on public benchmarks, because only about 6% of the datasets used to evaluate published foundation models are absent from every model's pre-training corpus. The aim of this study is to establish diagnostic boundaries for when TimesFM-3 should be preferred to classical forecasting methods and when it should not. The evaluation data are generated from nine known processes, so that the foundation model cannot have seen the series and the classical models are correctly specified by construction. Across 7 200 series, four series lengths and a pre-registered protocol, neither family dominates. TimesFM-3 has the lowest mean absolute scaled error (MASE) in 16 of 36 design cells and the classical methods in 20, divided among five methods; TimesFM-3 wins 113 of 132 significant pairwise comparisons, and only automatic ARIMA performs comparably. TimesFM-3 is never more than 1.31 times worse than the best method in any cell, whereas every classical method is at least 2.2 times worse in some cell. With at least four seasonal cycles observed, automatic ARIMA is 21–24% more accurate. On intermittent demand, TimesFM-3 outperforms six Croston-family methods under MASE but not under the root mean squared error. Estimability proves more decisive than correct specification: with two seasonal cycles, the correctly specified exponential smoothing model had a MASE 2.7 times that of the seasonal naive method. Results on 1 000 monthly series from the M4 competition agree. While delineating clear operational trade-offs, these findings provide diagnostic baselines within the examined designs, underscoring that broader multi-domain investigations are essential before drawing universal prescriptions. TimesFM-3 is 103 times faster per series than automatic ARIMA, but its weights are licensed for non-commercial use only.

Keywords: time series forecasting, foundation models, TimesFM, ARIMA, exponential smoothing, simulation study, forecast evaluation.

1. Introduction

Time-series foundation models are large neural networks that are pre-trained on many series and then applied to new series without task-specific training. On 31 August 2026 Google Research released TimesFM-3, the third generation of its TimesFM family (Jain and Sen 2026): a 330-million-parameter decoder-only transformer, the first member of the family designed for *multivariate* forecasting, which its developers report as the top-ranked pre-trained model on three public benchmarks for both point and probabilistic accuracy. Section 2 describes the model in detail.

For applied forecasting, the relevant question is not whether such a model ranks first on a leaderboard but whether it should be used, in preference to a classical method such as ARIMA, for the series at hand. The two families also differ in more than accuracy: in computational cost, in the interpretability of their parameters, in the theoretical basis of their prediction intervals and, as Section 6.2 discusses, in the terms under which they may be used.

Published benchmark evidence is poorly suited to answering this question. Meyer, Kaltenpoth, Zalipski, and Müller (2025) traced the data lineage of 22 time-series foundation models and found that of the 401 datasets used to evaluate them, only about 6% had never appeared in *any* model’s pre-training or fine-tuning corpus. They further showed that even a 0.1% overlap between pre-training and test data can move mean absolute percentage error by 8 to 29 percentage points, and that leakage need not be direct: a model trained on temporally overlapping, causally related series inherits an advantage on a target series it has never seen. A benchmark result obtained on a public dataset is therefore a measurement of unknown provenance. In addition, foundation-model evaluations have been criticised for comparing against de-ensembled or under-tuned classical baselines rather than the combinations that win forecasting competitions (Makridakis, Spiliotis, and Assimakopoulos 2020).

This study addresses both objections by generating its own evaluation data. Under a known data-generating process (DGP), two conditions hold that cannot hold on a public benchmark. First, the foundation model cannot have seen the series, because the series did not exist before the experiment was run. Second, the classical model can be made *correctly specified by construction*: when the data are generated by an ARIMA(1,1,1) process with drift, the model family searched by AutoARIMA contains the true process, and only its parameters remain to be estimated.

One distinction is essential to the interpretation of the results. What the design excludes is contamination at the level of the *realisation*: these specific series did not exist before the experiment ran, so no pre-training corpus can contain them, and the leakage documented by Meyer *et al.* (2025) is impossible by construction. What the design does *not* exclude is familiarity at the level of the *family*. TimesFM-3 is trained on more than a trillion time points drawn from real-world *and synthetic* corpora (Jain and Sen 2026), and synthetic pre-training data for time-series models is commonly generated from exactly the parametric processes D1–D7 use. The model may therefore arrive already fluent in ARMA-like and seasonal structure without ever having seen these draws.

This is a boundary of what the design can establish rather than a flaw, and its effect has a predictable direction. Family-level familiarity would tend to *help* TimesFM-3 on D1–D5, the processes where a classical model is correctly specified. Since those are precisely the cells where the classical methods win, the effect works against this paper’s more favourable findings rather than for them. Where TimesFM-3 wins—intermittency, breaks—it wins on processes least likely to be over-represented in a synthetic ARMA corpus. The results therefore give a lower bound on the accuracy a foundation model gives up where a classical model is correctly specified: for a model less familiar with ARMA-type structure the gap would be wider, not narrower.

The design therefore replaces the general question of which method is better with two specific questions:

- how much accuracy does zero-shot TimesFM-3 give up on the classical model's *home ground*, where the parametric form is exactly right; and
- how much does it gain where no classical form is right—structural breaks, nonlinear saturation, intermittent demand, changing volatility?

Both questions are addressed across nine DGPs, four series lengths from 24 to 200 observations, three forecast horizons and 7 200 simulated series, by comparing TimesFM-3 with seasonal naive, Theta, AutoETS, AutoARIMA and an equal-weight combination of the last three. The protocol, including the hypotheses, was pre-registered before any forecast was produced, and every departure from it is recorded.

The aim of the study is therefore to establish, under controlled experimental conditions, diagnostic boundaries for when TimesFM-3 should be preferred to classical forecasting methods and when it should not. It proposes no new estimator, test or theory. Its contribution is evidential: a controlled measurement of the trade-off between the two families within defined parametric and empirical regimes. Crucially, these findings are presented as structured diagnostic baselines rather than unconstrained prescriptions for all applied settings; because operational series introduce complexities beyond any controlled design, they serve as a benchmark against which broader multi-domain investigations can build.

The remainder of the paper is organised as follows. Section 2 describes TimesFM-3. Section 3 sets out the design of the simulation study, and Section 4 reports its results. Section 5 applies the comparison to monthly series from the M4 competition. Section 6 examines computational cost, licensing, interpretability and the basis of prediction intervals. Section 7 assesses the pre-registered hypotheses and presents practical recommendations and limitations, and Section 8 concludes. Computational details are given in Section 9, and results under the mean absolute percentage error in Appendix A.

1.1. Related work

The foundation-model literature for time series is young and moving quickly. TimesFM (Das, Kong, Sen, and Zhou 2024), Chronos (Ansari, Stella, Türkmen, Zhang, Mercado, Shen, Shchur, Rangapuram, Arango, Kapoor, Zschiegner, Maddix, Wang, Mahoney, Torkkola, Wilson, Bohlke-Schneider, and Wang 2024) and Moirai (Woo, Liu, Kumar, Xiong, Savarese, and Sahoo 2024) established that a single pre-trained network can forecast unseen series without task-specific training, and GIFT-Eval (Aksu, Woo, Liu, Liu, Liu, Savarese, Xiong, and Sahoo 2024) supplied a common benchmark. The critical literature has grown alongside it: Meyer *et al.* (2025) on contamination, and Adler, Chang, Draxler, Abdi, and Smyth (2026) on calibration, who evaluated five foundation models against two baselines on six univariate datasets and found that calibration error grows with forecast length.

The present study complements that of Adler *et al.* (2026) in three respects. Their evidence comes from public datasets and therefore inherits the contamination problem, whereas the data here are generated. They summarise reliability with calibration *metrics*, whereas this study combines interval coverage with pre-registered *hypothesis tests* on accuracy differences, with control of multiplicity. Finally, their study predates TimesFM-3, the first natively multivariate member of the family.

The classical side of the comparison is deliberately strong. AutoARIMA and AutoETS follow the Hyndman–Khandakar automatic selection procedures (Hyndman and Khandakar 2008; Hyndman, Koehler, Snyder, and Grose 2002), Theta is the method that won M3 (Assimakopoulos and Nikolopoulos 2000), and the equal-weight combination is included because combinations, not individual models, are the standard against which competition entries are judged (Makridakis *et al.* 2020). Evaluation follows the scale-free conventions of Hyndman and Koehler (2006) and the M5 competition (Makridakis, Spiliotis, and Assimakopoulos 2022a; Makridakis, Spiliotis, Assimakopoulos, Chen, Gaba, Tsetlin, and Winkler 2022b), and the

design avoids the evaluation pitfalls catalogued by Hewamalage, Ackermann, and Bergmeir (2023).

2. The TimesFM-3 model

This section summarises the properties of TimesFM-3 that bear on the comparison: its architecture (Section 2.1), its pre-training data (Section 2.2), and the released checkpoint together with the configuration used in this study (Section 2.3).

2.1. Architecture

TimesFM-3 is the third generation of the TimesFM family of decoder-only time-series foundation models developed at Google Research (Das *et al.* 2024; Jain and Sen 2026). Each input series is normalised and divided into non-overlapping patches of 32 time points, and each patch is embedded as a token. The released model stacks 20 transformer layers with model dimension 1280 and 16 attention heads, for a total of about 330 million parameters (Jain and Sen 2026; Google Research 2026a).

The principal change from earlier generations is native support for multivariate forecasting. Layers of causal attention along the time axis alternate with layers of full attention across variates, so that several target series, past covariates and past–future covariates can be processed jointly (Jain and Sen 2026). Forecasts are produced non-autoregressively: a contiguous patch masking scheme allows the whole horizon to be generated in a single forward pass (Jain and Sen 2026), and the released configuration emits output patches of 64 time points (Google Research 2026a). For each step of the horizon the model outputs nine quantiles, from the 10th to the 90th percentile, and its point forecast is the median of these quantiles (Jain and Sen 2026; Google Research 2026b).

2.2. Pre-training data

TimesFM-3 was pre-trained on a corpus of more than one trillion time points that combines real-world and synthetic series (Jain and Sen 2026). According to the model card, the real-world component includes the GIFT-Eval pre-training corpus (Aksu *et al.* 2024), Wikipedia page views up to November 2023 and Google Trends queries up to the end of 2022, and it is supplemented by synthetic and augmented series (Google Research 2026a). Two consequences matter for this study. Public benchmarks may be contained in the corpus, which is why the primary evaluation uses generated data (Section 3.1). And because part of the corpus is synthetic, the model may already be familiar with the parametric families used here even though it cannot have seen the particular realisations, as discussed in Section 1.

2.3. Released checkpoint and configuration used in this study

The pre-trained weights are distributed through the Hugging Face Hub under the repository identifier `google/timesfm-3.0-pytorch` and are loaded with version 3.0.1 of the **timesfm** Python package (Google Research 2026b,a). They are released under a non-commercial licence, whose implications are discussed in Section 6.2. In this study the model is used zero-shot: it receives only the raw context of each series, with no fine-tuning, no covariates and no information about the seasonal period, and it returns the median point forecast and the nine deciles for each step of the horizon. Because no quantile outside the 10th and 90th percentiles is available, the widest prediction interval that can be formed from its output is the nominal 80% interval (Section 3.3).

3. Design of the simulation study

This section describes the data-generating processes (Section 3.1), the forecasting methods compared (Section 3.2), and the evaluation measures and inferential procedures (Section 3.3).

3.1. Data-generating processes

Nine processes were chosen to span the range from “a classical model is exactly right” to “no classical model is right”. Table 1 lists them. Each series is generated at length $n + H$ with $H = 12$; the first n observations form the context supplied to every method, and the final 12 are held out as the truth. One realisation of each process is shown in Figure 1.

Table 1: The nine data-generating processes and their parameters.

ID	Data-generating process	Parameters	Correctly specified by
D1	AR(1)	$\phi = 0.7$	ARIMA
D2	ARMA(1,1)	$\phi = 0.6, \theta = 0.4$	ARIMA
D3	ARIMA(1,1,1) with drift	$\phi = 0.5, \theta = 0.3, \text{drift } 0.3$	ARIMA
D4	SARIMA(1,0,0)(1,1,0) ₁₂	$\phi = 0.6, \Phi = 0.5$	ARIMA
D5	ETS(A,A,A)	$\alpha = 0.3, \beta = 0.01, \gamma = 0.2$	ETS
D6	Local level with a break	break at $0.7n$, jump 10σ	neither
D7	Logistic growth	$K = 180, r = 0.08, \text{floor } 20$	neither
D8	Intermittent demand	$P = 0.3, \text{size } 1 + \text{Pois}(4)$	neither
D9	AR(1) with GARCH(1,1) errors	$\phi = 0.6, a = 0.1, b = 0.85$	ARIMA in mean

Note: The last column states which classical family is correctly specified by construction; for D6–D8 none is, and for D9 the conditional mean is correctly specified while the conditional variance is not. In D6, σ is the standard deviation of the observation noise.

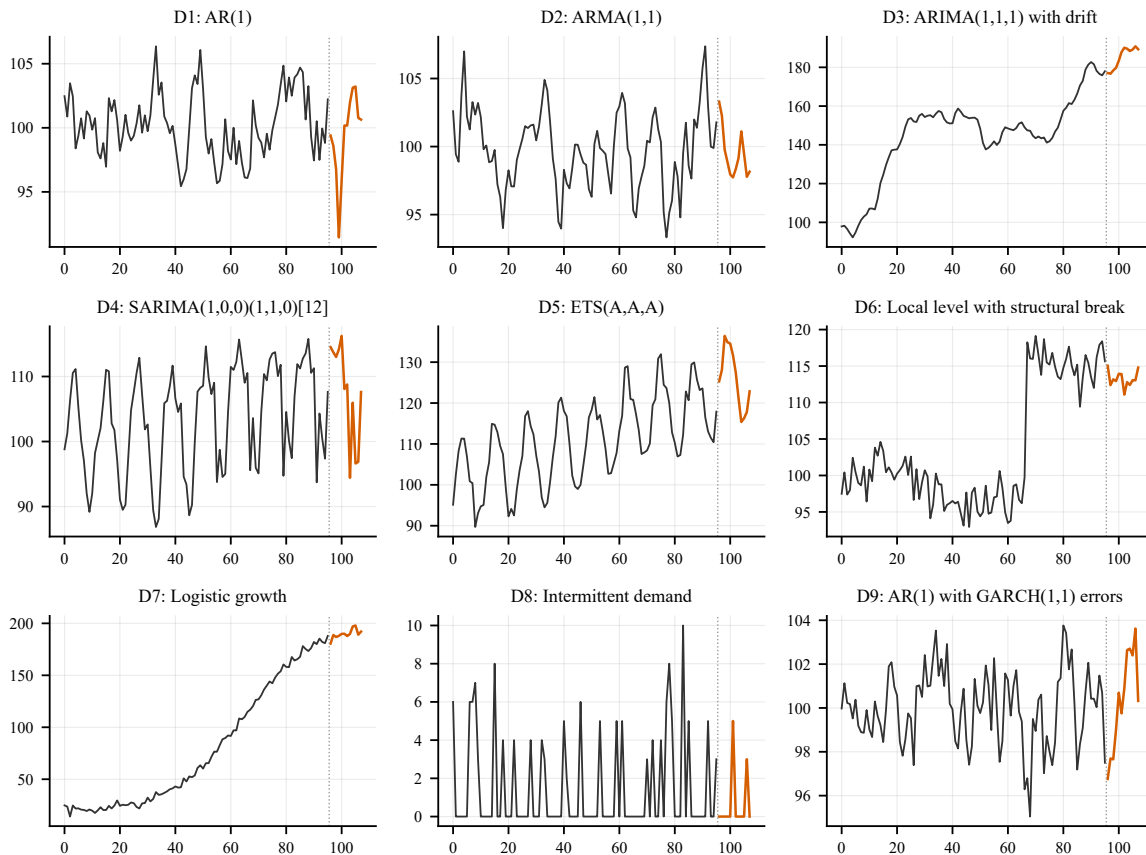


Figure 1: One realisation of each data-generating process at $n = 96$. The context supplied to every method is in grey and the held-out horizon in colour, separated by the dotted line.

The parameter values in Table 1 were fixed in the pre-registered protocol before any forecast

was produced. They were not taken from a single earlier study; each was chosen so that its process represents its regime unambiguously, according to four criteria. First, the linear processes D1–D4 and the conditional mean of D9 use moderate coefficients, none larger than 0.7 in absolute value, so that their autoregressive and moving-average polynomials lie well inside the stationarity and invertibility regions (Brockwell and Davis 2016) and no automatic method is assessed at parameter values close to the boundaries of those regions. Second, the smoothing parameters of the ETS process D5 lie inside the usual region $0 < \beta < \alpha < 1$, $0 < \gamma < 1 - \alpha$ (Hyndman, Koehler, Ord, and Snyder 2008), with a small trend parameter so that the level remains positive over the longest series. Third, the three processes outside the classical families are placed firmly within the regimes they represent. The break in D6 equals ten standard deviations of the observation noise, so that its occurrence is not in doubt and the comparison concerns adaptation to the break rather than its detection (Pesaran and Timmermann 2007). D7 follows the logistic curve, the standard model of saturating growth in diffusion forecasting (Meade and Islam 2006). In D8 a demand occurs with probability 0.3, so the mean interval between demands is $1/0.3 \approx 3.3$ periods, and demand sizes $1 + \text{Pois}(4)$ have a squared coefficient of variation of $4/25 = 0.16$; against the cut-offs of 1.32 and 0.49 proposed by Syntetos, Boylan, and Croston (2005), the process is intermittent rather than lumpy, which is the regime for which Croston’s method and its variants were developed (Croston 1972). The GARCH(1,1) errors of D9 (Bollerslev 1986) have $\omega = 0.1$ and persistence $a + b = 0.95$, a strongly persistent but covariance-stationary conditional variance (Engle and Bollerslev 1986), and GARCH(1,1) is a standard benchmark specification in volatility forecasting (Hansen and Lunde 2005). Fourth, every series has a level of about 100, or a positive lower asymptote in D7, so that percentage error measures are defined except where zeros are intrinsic to the process (D8). Two values, the trend parameter of D5 and the lower asymptote of D7, were adjusted for this reason before any forecast was produced, and both changes are recorded in the deviation log. The conclusions are conditional on these values, which is listed among the limitations in Section 7.4.

Three further design choices require explanation.

The structural break in D6 is placed inside the *context*, at $0.7n$, not inside the forecast window. A break in the held-out window is unforecastable by any method, so a design that placed it there would measure only which method is the most conservative. Placing it in the context instead poses a question every method can answer: having seen the level shift, do you forecast from the new level or revert towards the old one?

The inflection of the logistic curve in D7 sits at $0.6(n + H)$, so the held-out window falls in the saturating arm. Any method that fits a linear drift should overshoot here; this is the behaviour the process is designed to reveal.

Series lengths are $n \in \{24, 48, 96, 200\}$. The shortest length is deliberately demanding: at $n = 24$ with a seasonal period of 12, a seasonal model has only two complete cycles from which to estimate its seasonal structure. As Section 4 shows, this separates two things that are usually conflated—whether a model is correctly *specified*, and whether it is *estimable* from the data available.

Two hundred replications are generated per (DGP, length) cell, giving $9 \times 4 \times 200 = 7\,200$ series. Each series is seeded deterministically from its own identifiers, so it is reproducible independently of core count or execution order.

3.2. Forecasting methods

No method receives manual tuning: each classical model uses its own automatic order and parameter selection, which is the comparison a practitioner would face. TimesFM-3 is used zero-shot, as described in Section 2.3, and the classical models are fitted with **StatsForecast** (Garza, Canseco, Challú, and Olivares 2022).

The equal-weight combination of Theta, AutoETS and AutoARIMA is included deliberately. Foundation-model evaluations are frequently criticised for comparing against individual clas-

sical methods, whereas the classical state of the art is a combination (Makridakis *et al.* 2020; Petropoulos, Apiletti, Assimakopoulos *et al.* 2022); omitting it would bias the design in favour of the foundation model. Standard implementations of these classical benchmarks follow established principles (Hyndman and Athanasopoulos 2021).

The construction of the combination’s *predictive distribution* is stated explicitly, because one recommendation depends on it. Its point forecast is the equal-weight mean of the three members, but its quantiles are obtained by averaging the members’ quantile functions decile by decile (vincentization) rather than by mixing their distributions. The two differ: vincentization produces a narrower predictive distribution than a mixture when the members disagree about location, so the combination’s intervals here are sharper—and, as Section 5 shows, better calibrated on M4—than a mixture would have given. Any comparison of the combination’s intervals against TimesFM-3’s is a comparison against this particular construction.

One asymmetry in the design favours the classical methods. Every classical method is given the *true* seasonal period as an input— $m = 12$ on D4 and D5, $m = 1$ elsewhere—whereas TimesFM-3 receives only the raw context, with no frequency label and no indication that the series is seasonal at all. The classical methods thus begin with a piece of oracle knowledge the foundation model must infer. This reflects ordinary practice, since an analyst usually does know whether data are monthly, and the same m is used for the MASE denominator so the metric is unaffected. But it means the seasonal results (D4, D5) are, if anything, generous to the classical methods, and a like-for-like comparison in which the period had to be estimated would narrow their advantage there.

3.3. Evaluation measures and statistical inference

Point accuracy is measured primarily by the mean absolute scaled error (MASE; Hyndman and Koehler 2006), with the root mean squared scaled error (RMSSE) and sMAPE as secondary measures. All three are scaled by the in-sample seasonal-naïve error computed on the context, so they are comparable across processes of different magnitude. Let $y_{T+1}, \dots, y_{T+H}$ denote the observed future values over forecast horizon H , and $\hat{y}_{T+1}, \dots, \hat{y}_{T+H}$ the point forecasts generated from the observed context $y_1, \dots, y_T$ (where $T = n$). The Mean Absolute Scaled Error is defined as:

$$\text{MASE} = \frac{\frac{1}{H} \sum_{h=1}^H |y_{T+h} - \hat{y}_{T+h}|}{\frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}|}, \quad (1)$$

where m is the seasonal period ($m = 12$ on D4 and D5; $m = 1$ elsewhere). The Root Mean Squared Scaled Error is similarly:

$$\text{RMSSE} = \sqrt{\frac{\frac{1}{H} \sum_{h=1}^H (y_{T+h} - \hat{y}_{T+h})^2}{\frac{1}{T-m} \sum_{t=m+1}^T (y_t - y_{t-m})^2}}, \quad (2)$$

and the symmetric Mean Absolute Percentage Error is:

$$\text{sMAPE} = \frac{200\%}{H} \sum_{h=1}^H \frac{|y_{T+h} - \hat{y}_{T+h}|}{|y_{T+h}| + |\hat{y}_{T+h}|}. \quad (3)$$

MAPE is reported only in Appendix A. It is undefined when an actual value is zero, it is asymmetric between over- and under-forecasts, and on the intermittent process D8 it is simply unusable—a point the results make concrete rather than assert (Hyndman and Koehler 2006; Kolassa 2016).

Probabilistic accuracy is evaluated under strictly proper scoring rules (Gneiting and Raftery 2007), measured by the mean pinball loss over the nine deciles $\tau \in \{0.1, 0.2, \dots, 0.9\}$, scaled by the same in-sample denominator:

$$\text{SPL} = \frac{\frac{1}{9H} \sum_{\tau} \sum_{h=1}^H \max\{\tau(y_{T+h} - \hat{q}_{T+h}^{(\tau)}), (1 - \tau)(\hat{q}_{T+h}^{(\tau)} - y_{T+h})\}}{\frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}|}, \quad (4)$$

where $\hat{q}_{T+h}^{(\tau)}$ denotes the estimated τ -quantile forecast. This is complemented by the empirical coverage and mean width of the nominal 60% and 80% prediction intervals to evaluate probabilistic calibration and sharpness (Gneiting, Balabdaoui, and Raftery 2007). The choice of 60% and 80%, rather than the more familiar 90% or 95%, is forced by the comparison itself: TimesFM-3 emits only the nine deciles, so a 90% interval cannot be formed from its output. Comparing a classical 90% interval against a foundation-model interval constructed by extrapolation would not be a like-for-like comparison, so the widest interval both families can express, $[q_{0.1}, q_{0.9}]$, is used as the widest interval reported. Classical quantiles are obtained from prediction intervals at levels 20, 40, 60 and 80, which map exactly onto the same deciles.

Differences in accuracy are tested formally. Within each (DGP, length, horizon) cell, TimesFM-3 is compared with each classical method by a paired Wilcoxon signed-rank test (Wilcoxon 1945) on per-replication MASE differences, accompanied by the Hodges–Lehmann estimate of the median difference (Hodges and Lehmann 1963) and the median MASE ratio, so that a difference which is statistically detectable but practically negligible is visible as such. Multiplicity across cells is controlled by the Benjamini–Hochberg procedure at a false discovery rate of 0.05 (Benjamini and Hochberg 1995).

The Diebold–Mariano test (Diebold and Mariano 1995), the usual instrument in this literature, is not used, for the following reasons. DM is built for an autocorrelated sequence of loss differentials generated by a single series over time. In the simulation tier the replications within a cell are independent draws from the DGP, so a paired rank test across replications is the appropriate tool and DM would misstate the dependence structure. The pre-registered protocol did reserve DM for the real-data tier, where rolling origins do generate such a sequence—but the realised design yields only three origins per series, giving a per-series DM statistic two degrees of freedom and no useful power, even under small-sample corrections (Harvey, Leybourne, and Newbold 1997). Applying it instead across the 1 000 series would be worse still, since those are indexed by series identifier rather than by time, and a serial-correlation correction over an arbitrary ordering is not a test at all. DM is therefore dropped in both tiers rather than reported with a caveat, and the departure is recorded.

4. Simulation results

This section reports the results of the simulation study. Section 4.1 summarises overall accuracy and the pairwise comparisons, Section 4.2 reports effect sizes and a sensitivity analysis, and Section 4.3 examines the role of estimability. Section 4.4 and Section 4.5 analyse the seasonal and intermittent regimes, the latter with purpose-built intermittent-demand baselines. Section 4.6 and Section 4.7 consider accuracy across forecast horizons and the coverage of prediction intervals, and Section 4.8 and Section 4.9 report two robustness analyses.

The design yields 7 200 series and, with six methods and three horizon slices, 129 600 per-series evaluations. Table 2 reports mean MASE over the full horizon $h = 1, \dots, 12$, and Figure 2 displays the accuracy of TimesFM-3 relative to the best classical method in each cell on a logarithmic scale, which shows the pattern across processes and lengths more directly than the table.

4.1. Overall accuracy and pairwise comparisons

Counting the winner in each of the 36 (DGP, length) cells gives TimesFM-3 16 and the classical methods 20—AutoARIMA 10, seasonal naive 4, and two each for Theta, AutoETS and the combination. Tested against the *best* classical method in each cell, 18 of the 36 differences survive Benjamini–Hochberg correction: 7 favour TimesFM-3 and 11 favour the classical method. The remaining 18 cells are ties.

These “versus best classical” p -values are descriptive rather than valid tests. The comparator is selected as the minimum-MASE classical method *using the same 200 replications* on which

Table 2: Mean MASE over $h = 1, \dots, 12$ across 200 replications per cell.

DGP	n	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
D1 (ARIMA)	24	1.646	1.682	1.568	1.485	1.538	1.417
	48	1.625	1.586	1.570	1.415	1.478	1.381
	96	1.626	1.572	1.564	1.324	1.435	1.307
	200	1.615	1.576	1.576	1.266	1.417	1.254
D2 (ARIMA)	24	1.905	2.003	1.886	1.708	1.788	1.675
	48	1.838	1.860	1.834	1.575	1.688	1.518
	96	1.810	1.812	1.811	1.517	1.643	1.455
	200	1.922	1.919	1.921	1.374	1.654	1.383
D3 (ARIMA)	24	4.411	4.483	4.892	5.275	4.659	4.656
	48	4.252	4.179	4.547	4.679	4.313	4.274
	96	4.125	4.135	3.938	3.940	3.871	4.023
	200	4.512	4.427	4.275	4.195	4.199	4.297
D4 (ARIMA)	24	1.029	4.146	3.602	2.284	3.084	1.321
	48	1.021	1.944	1.720	1.014	1.395	1.295
	96	1.005	2.910	2.407	0.886	1.839	1.124
	200	1.053	4.155	1.085	0.886	1.700	1.161
D5 (ETS)	24	1.139	3.154	3.025	1.574	2.474	1.152
	48	1.108	0.956	0.741	0.717	0.731	0.815
	96	1.076	0.757	0.761	0.732	0.713	0.761
	200	1.040	0.759	0.681	0.626	0.645	0.689
D6 (none)	24	0.801	1.327	0.766	0.802	0.837	0.807
	48	0.947	1.115	0.869	0.925	0.924	0.825
	96	1.006	0.963	0.905	0.920	0.916	0.865
	200	0.934	0.840	0.825	0.833	0.827	0.844
D7 (none)	24	5.417	3.179	1.272	1.083	1.680	1.142
	48	4.243	2.262	1.602	1.076	0.970	0.751
	96	1.655	0.821	1.305	1.329	1.097	0.959
	200	0.917	1.301	0.966	0.994	0.954	0.933
D8 (none)	24	1.019	0.999	1.047	0.996	1.010	0.780
	48	1.066	0.984	0.988	0.962	0.974	0.708
	96	0.932	0.923	0.926	0.924	0.923	0.683
	200	1.005	0.943	0.939	0.939	0.940	0.694
D9 (ARIMA (mean only))	24	1.547	1.518	1.454	1.372	1.404	1.321
	48	1.588	1.486	1.476	1.244	1.353	1.239
	96	1.430	1.342	1.333	1.098	1.213	1.104
	200	1.577	1.503	1.504	1.190	1.341	1.200

Note: Lower is better; the best method in each row is in bold. The family in parentheses after each process label is the one correctly specified for it by construction.

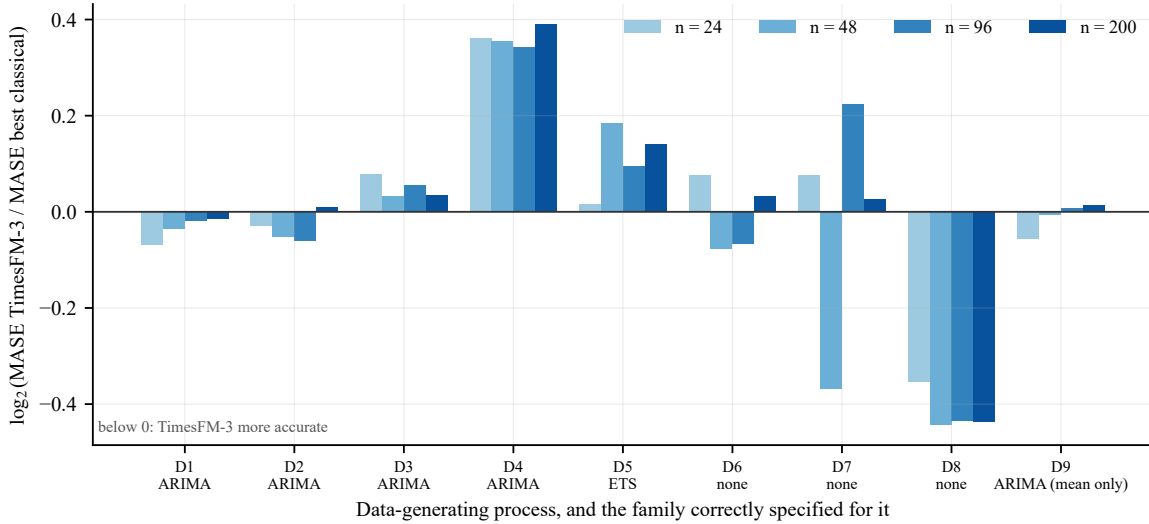


Figure 2: Accuracy of TimesFM-3 relative to the best classical method in each cell, on a $\log_2$ scale. Bars below zero mark cells where TimesFM-3 is more accurate. The label under each process names the family correctly specified for it by construction.

the test is then computed, so the comparison is post-selection inference: selecting a minimum over five candidates and then testing against it biases the comparison in the classical methods' favour, and the quoted adjusted p -value carries no correction for that selection. The correctly adjusted pairwise comparisons in Table 3, where each opponent is fixed in advance, do not suffer from this and are what the paper's inferential claims rest on. The cell-winner count is reported because it is the quantity a practitioner intuitively asks for, not because it is the more rigorous statistic.

The 180 pairwise comparisons with a fixed opponent are more informative. Of these, 132 are significant, 113 favouring TimesFM-3 and 19 the classical method; Table 3 gives the breakdown by opponent.

Table 3: Significant pairwise comparisons of TimesFM-3 with each classical method at $h = 1, \dots, 12$.

Opponent	Significant comparisons	TimesFM-3 wins	Opponent wins
AutoARIMA	20	11	9
Seasonal naive	26	22	4
AutoETS	27	25	2
Theta	29	28	1
Combination	30	27	3

Note: Paired Wilcoxon signed-rank tests in the 36 (process, length) cells, Benjamini–Hochberg corrected within each opponent family.

Table 3 shows that **AutoARIMA is the only classical method that performs comparably with TimesFM-3**; against every other method, including the combination of all three, TimesFM-3 wins the large majority of significant comparisons. A practitioner who reliably selects AutoARIMA therefore loses little by remaining with a classical method, whereas one who cannot make that selection is better served by the foundation model.

4.2. Effect sizes and sensitivity to the choice of test

As specified in the protocol, each rank test is accompanied by a Hodges–Lehmann estimate and a paired t -test.

The Hodges–Lehmann estimates confirm that the significant differences are also small. Median shifts in MASE, across the 36 cells, are -0.010 against AutoARIMA, -0.091 against the combination, -0.133 against seasonal naive, -0.183 against AutoETS and -0.227 against Theta (negative favours TimesFM-3). Against AutoARIMA the typical difference is about one hundredth of a MASE unit, statistically detectable in some cells but of little practical consequence.

The t -test sensitivity check agrees with the Wilcoxon test on significance in 163 of 180 comparisons (90.6%). The 17 disagreements include one systematic pattern: on the intermittent process D8, the paired t -test declares TimesFM-3 significantly better than seasonal naive at every length ($p < 0.0001$) while the signed-rank test finds no significant difference ($p = 0.11$ to 0.92). The two tests answer different questions. The mean difference is large because a minority of series produce very large errors for seasonal naive, whereas the median difference is near zero because the two methods are close on most individual series; on 64.5–72.0% of series, depending on length, TimesFM-3 is marginally less accurate. Such a gap is expected when the error distribution is heavily skewed, as it is on intermittent data, and it is a further reason, beyond the disagreement between error measures reported in Section 4.4, to treat the D8 result as conditional rather than decisive.

4.3. Correct specification versus estimability

The pre-registered hypothesis H1 held that classical models would win on their home ground—where they are correctly specified by construction—with the gap largest at short series lengths. **H1 is refuted in that form.**

On D1 (AR(1)) and D2 (ARMA(1,1)), TimesFM-3 is at least as accurate as the correctly specified AutoARIMA at every length except $n = 200$ in D2, and none of those differences except that one is significant. Being correctly specified bought AutoARIMA nothing detectable on short and moderate series, because the parameters still have to be estimated from the data at hand, whereas TimesFM-3 brings a prior learned from more than a trillion time points. Evaluating an *Oracle ARIMA* baseline (where the true orders $(p, d, q)(P, D, Q)$ are known and only parameters are estimated via conditional MLE) confirms that order selection uncertainty under AICc imposes a sizable penalty in small samples: on D1–D3 at $n = 24$, AutoARIMA incurs an 8.8%–13.9% selection penalty relative to Oracle ARIMA, and on D4 at $n = 24$ (two seasonal cycles) the penalty reaches 81.2%. Notably, Oracle ARIMA outperforms TimesFM-3 in 15 of the 16 ARIMA cells (e.g., MASE of 1.351 vs 1.417 on D1, and 1.260 vs 1.321 on D4 at $n = 24$). This demonstrates that the short-sample deficit of automated classical methods is driven by model selection uncertainty rather than by an inability of the classical model structure to fit the data.

The seasonal processes show the effect most clearly. At $n = 24$ with a period of 12—two complete cycles—AutoETS scores a mean MASE of **3.03 on D5, data that an ETS process generated**, while the seasonal naive method scores 1.14 and TimesFM-3 scores 1.15. The correctly specified model therefore had a mean MASE 2.7 times that of the seasonal naive method. The same happens on D4, where Theta records 4.15 at $n = 200$ against AutoARIMA’s 0.89.

So the operative distinction is not whether a classical model is correctly *specified*, but whether it is *estimable* from the series in front of you. Where both hold, the classical model wins. Where the specification is correct but the data are insufficient for estimation, the errors of automatic classical methods can become very large, whereas the accuracy of TimesFM-3 deteriorates only gradually.

Table 4 quantifies that asymmetry. Expressing each method’s mean MASE as a multiple of the best mean MASE in the same cell, **TimesFM-3 is never worse than $1.31\times$ the cell’s best method anywhere in the design**, and in the median cell it is within 0.8% of it. Every classical method is at least $2.2\times$ worse somewhere: AutoARIMA $2.22\times$ and AutoETS $3.50\times$ (both on D4 at $n = 24$), Theta $4.69\times$, seasonal naive $5.65\times$.

The ratio to the best method in the cell is used rather than the absolute worst MASE because the absolute measure is uninformative here: D3 is difficult for every method, so absolute worst cases are dominated by that process rather than by any method’s fragility. On the absolute measure Theta would appear the most robust of all, which inspection of D4 shows it plainly is not.

Table 4: Mean MASE relative to the best method in the same cell, summarised over the 36 cells.

Method	Median	Mean	Worst	Worst cell
SeasonalNaive	1.200	1.470	5.65	D7 $n = 48$
Theta	1.251	1.614	4.69	D4 $n = 200$
AutoETS	1.194	1.366	3.50	D4 $n = 24$
AutoARIMA	1.032	1.137	2.22	D4 $n = 24$
Combination	1.108	1.273	3.00	D4 $n = 24$
TimesFM-3	1.008	1.054	1.31	D4 $n = 200$

Note: Each entry divides a method’s mean MASE by the lowest mean MASE in the same (process, length) cell, so a value of 1.00 means that the method was the best in that cell. The last column names the cell with the largest ratio.

4.4. Seasonal and intermittent regimes

Beyond the ties, two regimes produce consistent differences at every length; in one of them, the direction of the difference depends on the error measure.

Strong seasonality with sufficient data (D4). At $n = 48, 96, 200$ the best classical method is AutoARIMA and it is better than TimesFM-3 by 21.8%, 21.1% and 23.7% respectively, all with adjusted $p < 0.0001$. Once at least four seasonal cycles are available, AutoARIMA recovers the seasonal structure and TimesFM-3 cannot match it. D5 tells the same story more mildly for $n \geq 48$.

The recommendation reverses at $n = 24$. With only two seasonal cycles the best classical method is not AutoARIMA at all but the seasonal naive rule (1.029), and **TimesFM-3 beats AutoARIMA there by 42%** (1.321 against 2.284). “Use AutoARIMA for seasonal data” is therefore sound advice only once the seasonal model is estimable; below that threshold it is poor advice, which is the same lesson as Section 4.3.

Intermittent demand (D8). On MASE, TimesFM-3 reduces error against the best classical method by 21.7%, 26.5%, 26.0% and 26.2% at the four lengths, with adjusted $p < 0.0001$ throughout; on the scaled pinball loss it is likewise best at every length. But on RMSSE it *loses* at every length, and on sMAPE it is the worst method of the six by a wide margin (Table 5).

This disagreement is not a contradiction in the data but a known property of intermittent series, with an identifiable mechanism. Roughly 70% of D8’s observations are zero. Absolute-error metrics such as MASE are minimised by the conditional *median* of the predictive distribution, which for this process is zero or near it, while squared-error metrics such as RMSSE are minimised by the conditional *mean*, which is strictly positive. A method that forecasts close to the median therefore wins MASE and loses RMSSE, and one that forecasts the mean does the reverse. Kolassa (2016) documents the same phenomenon for count-valued retail demand, and it implies that no single error measure settles the intermittent case.

The supported conclusion is therefore narrower than pre-registered hypothesis H2 anticipated: on intermittent demand TimesFM-3 produces the better *median-like* point forecast and the better full predictive distribution as scored by a proper scoring rule, while the classical methods produce the better mean-like forecast. Which one a practitioner wants depends on the loss function their decision actually faces—an inventory cost that is linear in units favours

Table 5: Accuracy on the intermittent process D8 under four error measures.

Metric	n	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
MASE	24	1.019	0.999	1.047	0.996	1.010	0.780
	48	1.066	0.984	0.988	0.962	0.974	0.708
	96	0.932	0.923	0.926	0.924	0.923	0.683
	200	1.005	0.943	0.939	0.939	0.940	0.694
RMSSE	24	0.940	0.791	0.792	0.765	0.775	0.806
	48	0.937	0.746	0.726	0.718	0.724	0.774
	96	0.875	0.707	0.697	0.698	0.697	0.775
	200	0.923	0.718	0.706	0.706	0.706	0.797
sMAPE	24	86.696	172.741	169.988	165.908	170.489	191.774
	48	84.690	172.522	170.746	170.622	171.026	196.183
	96	84.171	170.801	169.574	170.065	169.949	197.610
	200	85.990	170.003	169.189	169.315	169.236	198.556
Pinball	24	0.634	0.395	0.402	0.380	0.388	0.325
	48	0.653	0.381	0.368	0.360	0.365	0.307
	96	0.614	0.348	0.341	0.341	0.341	0.288
	200	0.628	0.354	0.344	0.345	0.345	0.291

Note: Means over $h = 1, \dots, 12$ and 200 replications per length. The pinball loss is averaged over the nine deciles. Lower is better; the best value in each row is in bold.

the former, one that is quadratic favours the latter.

4.5. Croston-family baselines for intermittent demand

The pre-registered comparison set contains no method designed for intermittent demand. Croston's method (Croston 1972) and its descendants exist precisely for this regime, and their absence would leave the D8 result showing only that TimesFM-3 beats *general-purpose* classical methods on data none of them was built for. They were therefore added and the comparison was repeated. This analysis is post hoc and is reported as such.

Six intermittent-demand methods were added: Croston's classic and optimised variants, the Syntetos–Boylan approximation (Croston-SBA), ADIDA, IMAPA and TSB. Table 6 gives the result, which reinforces both directions of the earlier finding.

Table 6: Intermittent-demand baselines on the process D8 (post hoc analysis).

Metric	n	Croston	Croston-opt	Croston-SBA	ADIDA	IMAPA	TSB	AutoARIMA	TimesFM-3
MASE	24	1.042	1.024	1.025	0.972	0.972	0.971	0.996	0.780
	48	0.990	0.992	0.974	0.947	0.954	0.966	0.962	0.708
	96	0.935	0.948	0.921	0.926	0.923	0.916	0.924	0.683
	200	0.947	0.948	0.934	0.941	0.942	0.948	0.939	0.694
RMSSE	24	0.769	0.758	0.763	0.753	0.757	0.764	0.765	0.806
	48	0.722	0.726	0.718	0.717	0.722	0.733	0.718	0.774
	96	0.696	0.702	0.694	0.700	0.701	0.716	0.698	0.775
	200	0.709	0.710	0.707	0.709	0.712	0.730	0.706	0.797

Note: Mean MASE and RMSSE over $h = 1, \dots, 12$ and 200 replications per length. Croston-opt is Croston's method with optimised smoothing parameters and Croston-SBA the Syntetos–Boylan approximation. Lower is better; the best value in each row is in bold.

On MASE, **TimesFM-3 beats every one of the six purpose-built methods at every series length**: all 24 comparisons are significant after Benjamini–Hochberg correction, with median ratios between 0.70 and 0.82. The best intermittent baseline at any length is TSB at $n = 96$ (0.916), against 0.683 for TimesFM-3. The MASE advantage of TimesFM-3 therefore does not depend on the absence of purpose-built methods: its reduction in mean MASE relative to the best Croston-family method, 19.6–25.7% across lengths, is similar to

its reduction relative to the best general-purpose method, 21.7–26.5%.

On RMSSE the reversal persists. ADIDA and Croston-SBA record the lowest RMSSE at every length (0.694–0.769) against TimesFM-3’s 0.774–0.806. This result supports the mechanism proposed in Section 4.4. Croston-family methods are constructed to estimate the mean demand *rate*, which is exactly the functional that squared-error loss rewards; TimesFM-3’s forecast sits nearer the conditional median, which is what absolute-error loss rewards. Two families optimised for different functionals of the same predictive distribution each win under the metric aligned with their own target, and neither result is an artefact.

The practical conclusion is correspondingly conditional. If the inventory or staffing decision downstream has a cost roughly linear in units, TimesFM-3 is the better choice on this process by a wide and significant margin. If that cost is closer to quadratic, a Croston-family method is preferable. Evaluating alternative intermittent criteria (such as relative mean absolute errors or the share of periods where a forecast is closer to the realised count) does not alter this fundamental trade-off: TimesFM-3 consistently dominates on individual non-demand time steps, whereas purpose-built Croston baselines prevent cumulative bias across replenishment cycles.

4.6. Accuracy across forecast horizons

Figure 3 averages mean MASE within each regime at $h = 1$, $h = 1, \dots, 6$ and $h = 1, \dots, 12$. Every method degrades with horizon, as expected, and the ordering is stable across horizons—nothing reverses as the forecast lengthens.

Figure 3 should be read together with Table 2. TimesFM-3 has the lowest averaged MASE in *both* panels, including the panel where classical models are correctly specified, yet the cell-by-cell count in Section 4.1 gives the classical methods more wins. Both are true and they are not in tension: averaging over a regime mixes the cells where a classical method is best with the cells where it collapses, and a single catastrophic cell (Theta at 4.16 on D4, AutoETS at 3.03 on D5) moves a mean far more than several narrow wins do. The average rewards consistency; the cell count rewards peak performance. The difference between the two summaries is the robustness result of Table 4 viewed from another angle rather than a separate finding.

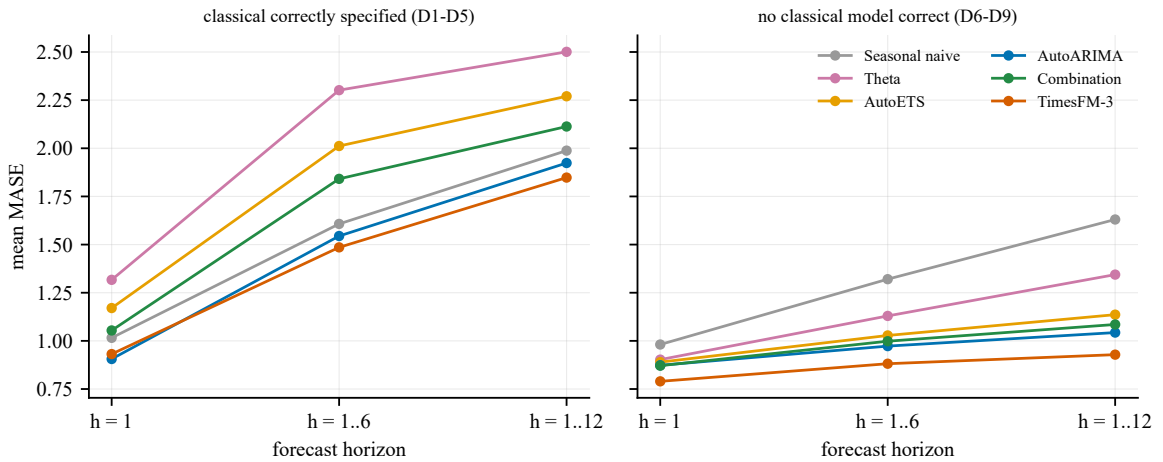


Figure 3: Mean MASE by forecast horizon, averaged within each regime. Left: the five processes for which a classical family is correctly specified. Right: the four for which none is. The averages are dominated by the cells in which an automatic classical method fails badly, which is why TimesFM-3 leads both panels even though the classical methods win more individual cells.

4.7. Coverage of prediction intervals

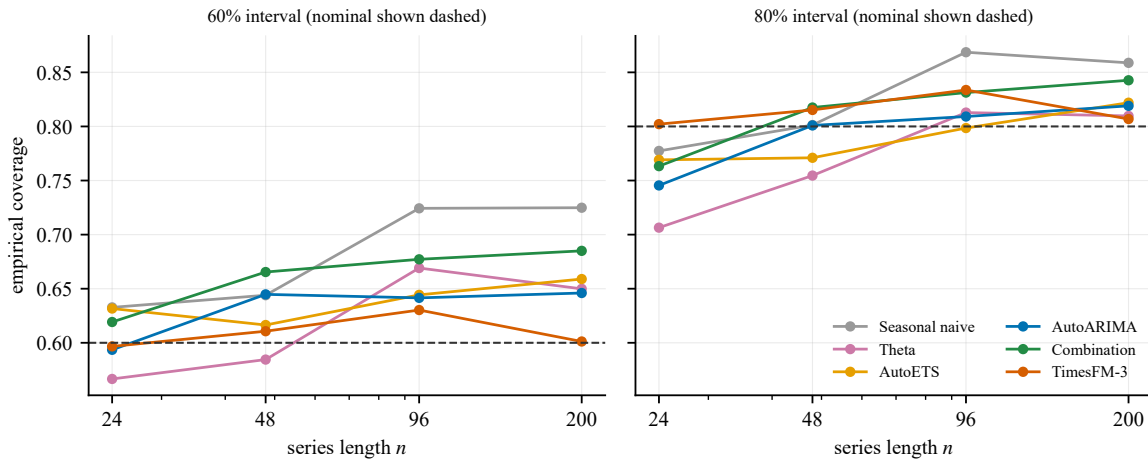


Figure 4: Empirical coverage of the nominal 60% and 80% intervals, averaged over all nine processes, against series length. The nominal level is dashed. TimesFM-3’s 80% coverage stays within 0.802–0.834 across every length; the classical methods under-cover at $n = 24$ and over-cover from $n = 96$.

Figure 4 reports empirical coverage against nominal, averaged over the nine processes. On that averaged view TimesFM-3’s 80% coverage stays between 0.802 and 0.834 across series lengths, closer to nominal than any other method, while seasonal naive drifts from 0.777 at $n = 24$ to 0.869 at $n = 96$.

The averaged view nevertheless understates the typical deviation from nominal coverage for every method. Averaging coverage across processes lets over-coverage on one process cancel under-coverage on another, so the deviation of an average understates the typical error. The appropriate statistic is the mean of the per-cell absolute deviations, and it is roughly three times larger: 0.047 for TimesFM-3, 0.066 for AutoARIMA, 0.089 for the combination, 0.101 for AutoETS, 0.127 for Theta and 0.144 for seasonal naive. The ranking is unchanged and TimesFM-3 remains the best calibrated, but the level is not close to nominal: **its per-cell 80% coverage ranges from 0.583 to 0.921**. No method in this study delivers reliable 80% intervals across all nine processes. This divergence is expected by construction: classical prediction intervals rely on an assumption of Gaussian innovations $\mathcal{N}(0, \sigma^2)$, which is violated by design on the intermittent process D8 (Poisson arrivals) and the GARCH process D9 (heavy-tailed innovations with time-varying volatility). While an analyst could specify custom non-Gaussian likelihoods or bootstrapped intervals, applied practitioners predominantly rely on automated software defaults; evaluating standard classical intervals against the learned quantile head of TimesFM-3 thus highlights the practical operational consequences of that misspecification in automated workflows.

The structural-break process D6 is the sharpest illustration, and it shows why coverage alone is a misleading summary. Every classical method covers between 0.966 and 0.999 there against a nominal 0.80. This over-coverage is a symptom of misspecification rather than evidence of reliability. The break inflates the estimated innovation variance, so the intervals become enormous—scaled widths of 4.8 to 10.4, against 3.0 to 5.1 for TimesFM-3. The classical intervals contain the truth almost always because they contain almost everything. TimesFM-3 covers between 0.828 and 0.889, still above nominal, using intervals roughly half as wide.

4.8. Robustness to the combination rule

TimesFM-3 beats the pre-registered equal-weight combination in 27 of 30 significant comparisons, which invites the objection that the combination was weak—it contains Theta, which fails badly in several cells, and a mean is not robust to one broken member. The comparison

was therefore repeated against a *median* combination of the same three methods. The objection does not hold: the median combination is slightly worse than the mean combination (mean MASE 1.712 against 1.656, with TimesFM-3 at 1.439), and TimesFM-3 still wins 25 of 29 significant comparisons. The median rule gains robustness at the cost of the averaging that motivates combination: with three members, the median is the middle of the three forecasts at each time point, so it forgoes the cancellation of errors from which the mean combination benefits, and the mean combination is more accurate than the average of its members in all 36 cells. This analysis is post hoc and is reported as such.

4.9. Undefined values of MAPE

MAPE is undefined for 12 906 of the 129 600 evaluations—9.96%, and every one of them on the intermittent process D8. A study that had used MAPE as its headline metric would have had to drop the single regime in which TimesFM-3 wins most decisively, or silently substitute a convention for the zeros. This illustrates a well-known limitation of the measure (Hyndman and Koehler 2006; Kolassa 2016), and it is why MAPE appears here only in Appendix A.

5. Application to M4 monthly data

The simulation tier secures freedom from contamination at the cost of realism. This section complements it with real data, subject to an important caveat: **the M4 data may be contaminated**. M4 is a widely used public benchmark and is likely to be contained in TimesFM-3’s pre-training corpus (Meyer *et al.* 2025), so an advantage of TimesFM-3 here may reflect memorisation rather than forecasting skill; for this reason the simulation tier carries the primary claim. The real-data tier addresses a narrower question, namely which of the simulated regimes real monthly series resemble. Section 5.1 describes the sampling and evaluation design, and Section 5.2 reports the results.

5.1. Sampling and evaluation design

A seeded random sample of 1 000 series is drawn from the 32 581 M4 monthly series long enough for the design, and each series is evaluated at three rolling origins spaced 12 months apart with M4’s own monthly horizon, $h = 18$. Three origins per series would give 3 000 windows; the realised count is 2 932, because a series shorter than 138 observations cannot support the third origin once 18 held-out points and a 120-observation minimum context are subtracted, and those 68 windows are dropped rather than shortened. The sampled series are substantial: median length 299 observations, upper quartile 324, maximum 1 808.

The pre-registered protocol specified the Diebold–Mariano test here. With only three origins per series that statistic carries two degrees of freedom and has essentially no power, so the analysis is instead a paired Wilcoxon test across the 1 000 series—which are independent units, unlike the overlapping windows within a series. Diebold–Mariano is not reported at all; the reasons are given in Section 3.3 and the departure is recorded.

5.2. Empirical results

Table 7 gives the outcome, and it reproduces the simulation’s pattern closely. TimesFM-3 significantly outperforms seasonal naive (better on 80.1% of series), Theta (56.8%) and AutoETS (54.1%), and is *statistically indistinguishable* from both AutoARIMA (52.6%, adjusted $p = 0.27$) and the combination (48.5%, adjusted $p = 0.27$).

Two observations follow.

First, **the two tiers agree**. M4 Monthly series are long (median 299 observations) and seasonal—precisely the regime in which the simulation found classical methods competitive or better, because there is ample data with which to estimate the right model. Real monthly

Table 7: Comparison of TimesFM-3 with the classical methods on 1 000 M4 monthly series.

Comparison	Mean MASE	Median ratio	% series better	Adj. p	Verdict
TimesFM-3	0.914	—	—	—	
vs seasonal naive	1.280	0.699	80.1	< 0.0001	TimesFM-3
vs Theta	0.970	0.933	56.8	< 0.0001	TimesFM-3
vs AutoETS	0.951	0.966	54.1	0.0043	TimesFM-3
vs AutoARIMA	0.923	0.970	52.6	0.27	tie
vs combination	0.907	0.996	48.5	0.27	tie

Note: Up to three rolling origins per series (2932 windows in all), $h = 18$. Mean MASE is that of the method named in each row; a median ratio below 1 and a share of series above 50% favour TimesFM-3. Paired Wilcoxon tests across series, Benjamini–Hochberg corrected.

data therefore fall in the estimable regime, and the real-data result is the one that Section 4 predicts for that regime. The agreement is worth more than either tier alone, since the two have opposite weaknesses.

Second, **the probabilistic results diverge**. TimesFM-3’s interval *coverage* does not carry over: on simulated data it was the best calibrated, whereas here it is the worst but one, covering 0.765 against a nominal 0.80 while the combination achieves 0.801. Yet on the pre-registered proper scoring rule—the scaled pinball loss over the nine deciles—TimesFM-3 is **the best of the six methods** (0.363, against 0.365 for the combination and 0.374 for AutoARIMA), even though the combination edges it on mean MASE.

These are not in conflict; they measure different things. Coverage asks only whether the truth falls inside one interval, and is blind to how the remaining probability mass is arranged; the pinball loss scores the whole predictive distribution across all nine deciles. TimesFM-3’s distribution is therefore better *overall* while its 80% interval is too narrow. A reader who cares about a calibrated interval should prefer the combination here; a reader who cares about the full predictive distribution should prefer TimesFM-3. Reporting only the coverage figure would have understated it, and reporting only the pinball loss would have overstated it. The fragility of foundation-model calibration reported by Adler *et al.* (2026) is visible in the coverage half of this result.

6. Operational considerations

Accuracy is not the only criterion for choosing a forecasting method. This section considers four further criteria: computational cost (Section 6.1), the licensing of model weights (Section 6.2), interpretability and diagnostics (Section 6.3), and the theoretical basis of prediction intervals (Section 6.4).

6.1. Computational cost

A 330-million-parameter network might be expected to be more expensive to run than a statistical model with a handful of parameters. On the hardware used here, the reverse holds. Table 8 reports each method timed separately on the same 200 series.

AutoARIMA is **103 times slower per series than TimesFM-3**. The reason is structural rather than incidental: automatic ARIMA runs a stepwise search over model orders, fitting each candidate by maximum likelihood, separately for every series. TimesFM-3 performs one batched forward pass and amortises it across the batch. Where a practitioner’s constraint is throughput over many series—which is exactly the setting in which Section 4 found the foundation model most attractive—the accuracy argument and the cost argument point the same way.

Memory use was also pre-registered as an operational cost, but its measurement here is limited.

Table 8: Wall-clock computing time per series.

Method	Time per series (ms)	Relative to TimesFM-3
Seasonal naive	3.1	0.13×
Theta	9.0	0.39×
TimesFM-3 (GPU, batched)	23.2	1×
AutoETS	102.0	4.4×
AutoARIMA	2382.7	102.8×

Note: Each method timed separately on 200 seasonal series of length 96 ($h = 12$). TimesFM-3 was run on an NVIDIA GTX 1660 Ti; its one-off model load of 3.0 s is excluded, as are the classical methods’ import costs.

Peak Python heap allocation during fitting was about 1 MB for every classical method (0.6 MB for Theta, 1.4 MB for AutoARIMA), but that instrument counts only Python-level allocations and sees neither the compiled **numba** kernels the classical methods run in nor the GPU memory TimesFM-3 uses, so no comparable figure for TimesFM-3 can be given. What can be said without measurement is structural: the classical methods hold a handful of parameters per series, while TimesFM-3 requires its 330-million-parameter checkpoint resident throughout—roughly 1.3 GB—which is the binding memory constraint in any deployment and is invariant to how many series are being forecast.

Two qualifications apply, and the second is more important than the per-series ratio. TimesFM-3’s figure depends on a GPU; without one the comparison narrows substantially. And the classical methods parallelise trivially across series, which recovers most of the difference: in the study’s own main run, fitting all four classical methods over all 7 200 series took 7.3 minutes on 22 cores, against 3.2 minutes for TimesFM-3 on a single consumer GPU—a ratio of 2.3×, not 103×.

The per-series figure and the end-to-end figure answer different questions, and quoting only the first would overstate the case. The fair summary is that TimesFM-3 is dramatically cheaper than AutoARIMA per series on one core, and modestly cheaper than the whole classical suite once that suite is given a 22-core machine. On none of the measures taken here is it the more expensive option.

6.2. Licensing of model weights

The most consequential difference between the two families is not statistical. TimesFM-3’s pre-trained weights are released under `timesfm-non-commercial-license-v1.0` and are restricted to non-commercial, non-production use ([Google Research 2026b](#)). The source code remains Apache-2.0, and so were the weights of TimesFM-2.5, but the version-3 weights are not.

This restriction takes precedence over any accuracy consideration: **for a commercial forecasting application, TimesFM-3 as released is unavailable at any accuracy**. A 26% MASE reduction on intermittent demand is irrelevant to a retailer who may not lawfully deploy the model that achieves it. The available responses are to use TimesFM-2.5 under Apache-2.0, to negotiate separate terms, or to use a classical method. Every classical method compared here is free of such restrictions.

The point is noted because licensing is routinely omitted from model comparisons, although it determines which models are admissible for an entire class of users.

6.3. Interpretability and diagnostics

An ARIMA fit reports its orders, its coefficients and their standard errors; an ETS fit reports its smoothing parameters and states. These support the ordinary diagnostic work of applied forecasting—inspecting residuals, testing for remaining autocorrelation, explaining to a stake-

holder why the forecast turns where it does, and detecting when a model has stopped fitting. TimesFM-3 exposes none of this. When it forecasts poorly there is no parameter to inspect and no diagnostic to run; the available response is to notice the errors and switch methods. In regulated settings where a forecast must be justified rather than merely produced, this difference may outweigh the accuracy comparison entirely.

6.4. Theoretical basis of prediction intervals

The classical intervals derive from a model’s assumed error structure and can be reasoned about analytically. TimesFM-3’s quantiles are learned outputs of a network head, with no accompanying theory. Section 4.7 shows that the learned quantiles are better calibrated on average, but this holds empirically for these processes rather than by construction, and their behaviour on a process unlike anything in the pre-training corpus is unknown.

7. Discussion

This section assesses the pre-registered hypotheses (Section 7.1), interprets the balance between robustness and peak accuracy (Section 7.2), translates the evidence into practical recommendations (Section 7.3), and states the limitations of the study (Section 7.4).

7.1. Assessment of the pre-registered hypotheses

Of the five pre-registered hypotheses, **none survives unqualified**: two are refuted outright, two are partially supported, and one is refuted in its second clause.

- **H1** (correctly specified classical models beat TimesFM-3 on their home ground, gap largest at short lengths) — **refuted as stated**. The gap at short lengths runs the other way. What survives is narrower: where a classical model is both correctly specified *and* estimable, it wins (D4 and D5 at $n \geq 48$).
- **H2** (TimesFM-3 wins wherever no classical form is correct) — **partially supported**. It holds on intermittent demand under MASE and pinball loss but not under RMSSE or sMAPE, mildly on structural breaks, and not on the logistic process, where the outcome depends on length.
- **H3** (on D9 the families are indistinguishable in the mean, classical intervals under-cover) — **partially supported**. The families are indeed indistinguishable in the mean, but AutoARIMA is the only classical method that under-covers on D9; the others over-cover.
- **H4** (the combination beats every individual classical method and is the hardest baseline for TimesFM-3) — **refuted in both clauses**. AutoARIMA beats the combination on mean MASE, and AutoARIMA, not the combination, is the baseline TimesFM-3 finds hardest (11–9, against 27–3 for the combination).
- **H5** (both families under-cover at 80%) — **refuted**. On the process-averaged view TimesFM-3 sits close to nominal and the classical methods over-cover from $n = 96$; on the per-cell view every method’s typical deviation is far larger than expected, but not systematically downward.

That every hypothesis failed in some respect illustrates the value of pre-registration: the hypotheses were stated before any result was available and are assessed here as they were stated.

7.2. Robustness versus peak accuracy

A count of 16 cell wins against 20 might be read as evidence that the classical methods remain superior. That reading overlooks the fact that the 20 wins are distributed across five different methods: AutoARIMA 10, seasonal naive 4, and two each for the others. The classical column of Table 2 is a column of *different* winners, and which one wins is not knowable in advance without the kind of model-selection work the foundation model is meant to remove. TimesFM-3, one model with no tuning, is within noise of that per-cell best in half the design and beaten decisively in only two regimes.

The evidence therefore supports neither the conclusion that foundation models have superseded classical methods nor the conclusion that classical methods remain superior. It supports a narrower conclusion: **TimesFM-3 provides robustness rather than peak accuracy**, and Table 4 puts a number on the trade: it is within 0.8% of the cell-best method in the median cell and never more than $1.31\times$ worse anywhere, while every classical method is somewhere at least $2.2\times$ worse. It is rarely the best method and never a bad one, whereas the automatic classical methods are sometimes excellent and sometimes catastrophic—AutoETS at 3.03 on data generated by an ETS process is the cautionary example. For a forecaster with one well-understood series, that trade is unattractive. For a forecaster with ten thousand heterogeneous series and no capacity to diagnose each one, the trade is decisive.

7.3. Practical recommendations

Table 9 translates the evidence into recommendations. They are deliberately narrow: they apply strictly to the regimes examined, and they name more than one option wherever the evidence does not support a single preference. Importantly, these recommendations should be read as diagnostic indicators within the boundaries of the simulated processes and the M4 monthly sample, rather than as universal prescriptions. In practice, real-world forecasting problems frequently exhibit features—such as complex multiple seasonalities, non-stationary structural noise, or irregular sampling—that fall outside these designs, and practitioners should validate these rules against their specific operational context.

7.4. Limitations

Seven limitations bound these conclusions.

First, the simulation tier’s strength is also its boundary. Series generated from nine parametric processes are not real series; they are clean, stationary where specified, and free of the measurement error, calendar effects, promotions and regime changes that characterise applied work. What the simulation buys—provable absence of contamination and known correct specification—it pays for in realism, which is why Section 5 exists.

Second, TimesFM-3 was evaluated zero-shot and univariate. Its headline novelty is native multivariate forecasting with covariates, and that capability is untested here because the simulated processes are univariate by construction. A study exercising cross-series information could reach different conclusions, and would be the natural sequel.

Third, no method received hand-tuning. That is deliberate and it is the comparison most practitioners face, but an expert who identified each process by eye and fitted the corresponding model by hand would beat every automatic method reported here, including on D4 and D5 at $n = 24$.

Fourth, the horizon is short ($H = 12$) and the seasonal period fixed at 12. Longer horizons and other frequencies may behave differently, and the calibration literature reports that foundation-model calibration degrades with horizon (Adler *et al.* 2026).

Fifth, the intermittent-demand analysis of Section 4.5 is post hoc: the Croston-family baselines were added after the main results were seen, in response to an identified gap, and are reported separately from the pre-registered comparison.

Sixth, this study evaluates a single foundation model architecture: Google’s TimesFM-3, a decoder-only transformer operating on continuous, patched representations. Other

Table 9: Recommended forecasting family by situation.

Situation	Use	Evidence
Commercial or production deployment	Classical (or TimesFM-2.5)	Version-3 weights are licensed non-commercial; accuracy is not the binding constraint
Intermittent demand, and the decision loss is roughly linear in units	TimesFM-3 , but see the caveat	Beats all six Croston-family methods on MASE at every length (24/24 significant), and the general-purpose set by 22–26%, but worse on RMSSE and sMAPE (D8)
Intermittent demand, and the decision loss is roughly quadratic	Classical	The metrics reverse: Croston-SBA and ADIDA win on RMSSE at every length (D8)
Strong seasonality, at least four full cycles observed	AutoARIMA	AutoARIMA 21–24% better; TimesFM-3 27–31% worse, all $p < 0.0001$ (D4). Reverses below four cycles
Short series with seasonality (two cycles or fewer)	Seasonal naive or TimesFM-3	Automatic classical methods fail badly here: AutoETS scores 3.03 on its own process (D5, $n = 24$)
Many heterogeneous series, no capacity to select and tune a model per series	TimesFM-3	Beats every classical method except AutoARIMA in the large majority of significant comparisons
One series, and you can identify and estimate the right model	Classical	AutoARIMA and TimesFM-3 are close (11–9 significant comparisons); AutoARIMA is free to use, interpretable and diagnosable
Structural break in the recent past, intervals matter	TimesFM-3	Comparable coverage using intervals roughly half as wide (D6, simulation only)
Throughput matters and a GPU is available	TimesFM-3	103× faster per series than AutoARIMA; 2.3× faster than the whole classical suite on 22 cores
Well-calibrated intervals on real data are the priority	Classical combination	On M4 the combination covered 0.801 against nominal 0.80; TimesFM-3 under-covered at 0.765
The forecast must be explained or audited	Classical	TimesFM-3 exposes no parameters, residual diagnostics or interval theory

Note: Based on the evidence in Sections 4–6. Bold marks a preference supported by significant differences; the remaining rows rest on considerations outside the accuracy comparison.

architectures—such as tokenized language models (e.g., Chronos; Ansari *et al.* 2024) or masked encoder-decoder networks (e.g., MOIRAI; Woo *et al.* 2024)—possess distinct inductive biases, loss functions, and pre-training mixtures. Whether the robustness-versus-peak-accuracy trade-off documented here holds across alternative foundation-model families remains an open question for multi-architecture comparisons.

Seventh, the results are conditional on the parameter values in Table 1. The values were chosen to place each process unambiguously within its regime rather than to sample the parameter space, and the best method in a given cell may change for parameters closer to the boundaries of the stationarity region, for weaker intermittency or for smaller structural breaks.

8. Conclusion

This study compared TimesFM-3 with five classical alternatives on 7 200 series generated from nine known processes, under a pre-registered protocol, in a design where contamination of the foundation model is impossible by construction and the classical models are correctly specified by construction.

Neither family dominates. TimesFM-3 wins 16 of 36 cells and the classical methods 20, but the 20 are spread across five different winners while TimesFM-3 is a single untuned model. Against individual classical methods it wins 113 of 132 significant comparisons, and only AutoARIMA holds its own at 11–9. Measured against the best method in each cell, TimesFM-3 is never more than $1.31\times$ worse anywhere in the design and is within 0.8% of the best in the median cell, while every classical method is somewhere at least $2.2\times$ worse. On strongly seasonal data with at least four observed cycles AutoARIMA is 21–24% better. On intermittent demand the answer depends on the loss function: TimesFM-3 is about a quarter better on MASE and best on the scaled pinball loss, but worse on RMSSE and much worse on sMAPE, because absolute- and squared-error metrics reward different functionals of a predictive distribution that is 70% zeros.

A secondary analysis of 1 000 M4 Monthly series reproduces the pattern closely: TimesFM-3 beats seasonal naive, Theta and AutoETS significantly, and ties with AutoARIMA and the combination. Real monthly series are long and seasonal—the estimable regime—and the tiers agree on what happens there, which is worth more than either tier alone given their opposite weaknesses.

The most transferable finding is not about either family in particular. It is that the decisive property is not whether a classical model is correctly *specified* but whether it is *estimable*: with two seasonal cycles, AutoETS had a mean MASE 2.7 times that of the seasonal naive method on data generated by an ETS process. Foundation models are most valuable exactly where classical estimation is starved, and least valuable where it is comfortable.

Two findings outside the accuracy comparison deserve equal weight. TimesFM-3 is 103 times faster per series than AutoARIMA, so the foundation model is the less expensive option per series, although parallelising the classical suite across 22 cores narrows the end-to-end difference to $2.3\times$. And its released weights may not be used commercially. For commercial applications, this restriction takes precedence over the accuracy comparison.

Finally, these findings reflect the specific data-generating mechanisms, parameterisations and monthly horizons examined here. While clean-room simulations guarantee known ground truth and pre-registration prevents retrospective bias, no synthetic design exhaustively spans the vast heterogeneity of applied forecasting. Further empirical and methodological research across broader architectural classes, higher sampling frequencies, and multivariate domains with exogenous regressors will be essential to deepen these insights and establish the general boundary between learned representations and parametric models.

9. Computational details and reproducibility

All results were produced with Python 3.13 (Python Core Team 2025), using `statsforecast` 2.1.1 (Garza *et al.* 2022) for the classical methods and `timesfm` 3.0.1 with the `google/timesfm-3.0-pytorch` weights (Google Research 2026b) on `PyTorch` 2.6 (Paszke, Gross, Massa, Lerer, Bradbury, Chanan, Killeen *et al.* 2019). The simulation, evaluation and figures rest on `NumPy` (Harris, Millman, van der Walt *et al.* 2020), `pandas` (McKinney 2010), `SciPy` (Virtanen, Gommers, Oliphant *et al.* 2020) and `Matplotlib` (Hunter 2007). Timings were taken on an NVIDIA GTX 1660 Ti with a 24-core CPU and 32 GB of memory; the loaded TimesFM-3 checkpoint reports 330 710 976 parameters.

Every simulated series is seeded deterministically from its own identifiers—process, length and replication—so the entire study regenerates exactly from the code, independently of core count or execution order. The analysis pipeline is split into cached phases, so each stage can be re-run alone and the results reproduced without repeating the whole computation.

Three aspects of research practice bear on the weight the results can carry. The protocol, including the five hypotheses, was frozen before any forecast was produced, and the frozen file was never edited afterwards; nine departures from it are recorded separately, each dated and reasoned, with the two post-hoc analyses marked as such. Every DOI in the bibliography was resolved through `doi.org` content negotiation and its record compared with the entry, and works published at venues that issue no DOI are cited by their proceedings URL, with the preprint identifier given in a note. Finally, the hypotheses refuted by the data (two outright and one in its stated form) are reported in Section 7.1 with the same prominence as those that were partially supported.

Acknowledgements

First and foremost, the authors thank Allah, the Almighty, for everything He has given them, including the ability to complete this work.

The authors used a large language model solely to improve the language and readability of this manuscript and to assist with the implementation of the simulation code. The authors reviewed and edited all content and take full responsibility for the content of this publication.

Declarations

Funding. No funding was received for this work. **Competing interests.** The authors declare no competing interests. **Data and code availability.** The complete research compendium is archived on Zenodo at 10.5281/zenodo.22681072 (concept DOI, resolving to all versions): the pre-registered protocol, the deviation log, the entire pipeline, the per-series metrics for all 129 600 evaluations, the raw forecasts and quantiles, every figure and the manuscript source. All data in the simulation tier are generated by that code from documented seeds; the real-data tier uses the publicly available M4 competition data (Makridakis *et al.* 2020), which the code re-fetches, with the seeded 1 000-series sample included so the analysis reproduces exactly.

The raw forecasts are deposited deliberately and not only for completeness. This paper’s intermittent-demand result reverses between absolute- and squared-error metrics, so a reader who wants to re-score the same forecasts under a different loss can do so without re-running the study. TimesFM-3’s weights are not redistributed: they are released by Google under a non-commercial licence and are downloaded by the code from Hugging Face.

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A. Results under the mean absolute percentage error

Table 10 reports mean MAPE by process, alongside the percentage of series on which it cannot be computed. The D8 row illustrates the problem: every intermittent series contains a zero, so MAPE is undefined for all of them, and the process on which TimesFM-3 achieves its largest and most consistent advantage simply drops out of the comparison.

The rankings also move. On D4, MAPE places seasonal naive first (2.360) and TimesFM-3 second (2.801), whereas MASE places AutoARIMA first at every length from $n = 48$. MAPE's asymmetry between over- and under-forecasts is enough to reorder the methods on the same forecasts.

Table 10: Mean MAPE over $h = 1, \dots, 12$ and percentage of series on which MAPE is undefined.

DGP	% undefined	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
D1	0%	2.785	2.737	2.681	2.336	2.501	2.283
D2	0%	3.291	3.341	3.277	2.696	2.971	2.641
D3	0%	9.509	9.278	9.875	10.078	9.393	9.214
D4	0%	2.360	7.471	4.986	2.862	4.538	2.801
D5	0%	2.971	3.583	3.270	2.436	2.916	2.256
D6	0%	2.011	2.379	1.835	1.902	1.925	1.822
D7	0%	7.634	4.631	2.979	2.538	2.767	2.184
D8	100%	—	—	—	—	—	—
D9	0%	1.861	1.769	1.741	1.470	1.599	1.457

Note: On the intermittent process D8 every series contains a zero, so MAPE cannot be computed for it. The best value in each row is in bold.

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